



DSA in C++

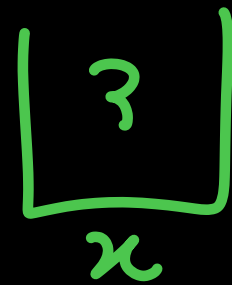
Conditionals

Lecture 05

Range of int

int x;

$$-2^{31} \leq x \leq 2^{31} - 1$$



short → 16 bits



int
short
long long

integers

bit → 0 or 1

int → 32 bits → 2^{32} numbers

↓

2

Range of int

1 byte = 8 bits

int x ;

char → 1 byte → 8 bits

int → 4 bytes = 32 bits

long long → 8 bytes = 64 bits

short → 2 bytes = 16 bits

float → 4 bytes

double → 8 bytes

bool = 1 byte

double, short, long long

long long x ; $-2^{63} \leq x \leq 2^{63} - 1$

short y , $-2^{15} \leq y \leq 2^{15} - 1$

double
float] real no.s
more precise

Multiple Conditions using `&&` and `||`

↓
logical
and

intersection

↓
logical
or

union

Ques: Take positive integer input and tell if it is a four digit number or not.

✗ `if (1000 ≤ n ≤ 9999) cout << "4 digit no";`

✓ `if ( n ≥ 1000 and n ≤ 9999 )`
 c1 c2

Ques: Take positive integer input and tell if it is a four digit number or not.

```
✓int n;  
✓cout<<"Enter a number: ";  
✓cin>>n;  
✓if(n >= 1000 && n <= 9999) cout<<"4 Digit number";  
else cout<<"not a 4 digit number";
```

Output/ Input

Enter a no. 12121

not a 4 digit no

(12121)
n

HW: Take integer input and tell if its magnitude is smaller than 69 or not.

$$|n| < 69$$

Ques: Take positive integer input and tell if it is divisible by 5 or 3.

$$n = 3$$

✓

$$n = 9$$

✓

$$n = 25$$

✓

$$n = 18$$

✓

$$n = 7$$

✗

$\text{if}(\underbrace{n \% 5 == 0}_{c1} \text{ or } \underbrace{n \% 3 == 0}_{c2})$

Ques: Take positive integer input and tell if it is divisible by 5 or 3.

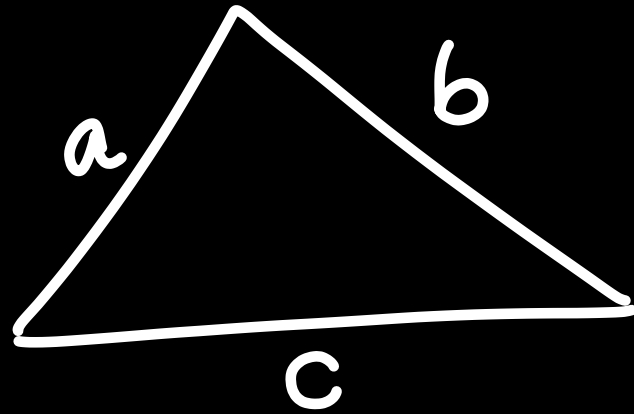
```
✓int n;  
✓cout<<"Enter a number: ";  
✓cin>>n;  
if(n%5 == 0 || n%3 == 0) cout<<"Divisible by 5 or 3";  
else cout<<"Condition not met";
```

1/0

Enter a number . 9

13
n

Ques: Take 3 positive integers input and tell if they can be the sides of a triangle or not.



if $(a+b > c \ \& \ a+c > b \ \& \ b+c > a)$

8 12 4 α

Ques: Capital Alphabet or not

take 'char' input & tell if it is -

'\$' → X

'B' → ✓

'c' → ✗

'k' → ✗

'A' → 65

.

if(65 ≤ (int)ch ≤ 90) ✓

else ✗

'Z' → 90

Ques: Take 3 positive integers input and print the greatest of them. a, b, c

if($a > b$ ~~and~~ $a > c$) a sabse bada hai

else if($b > a$ ~~and~~ $b > c$) b —

else c —

HW: Take 3 positive integers input and print the least of them.

Ques: Take positive integer input and print:

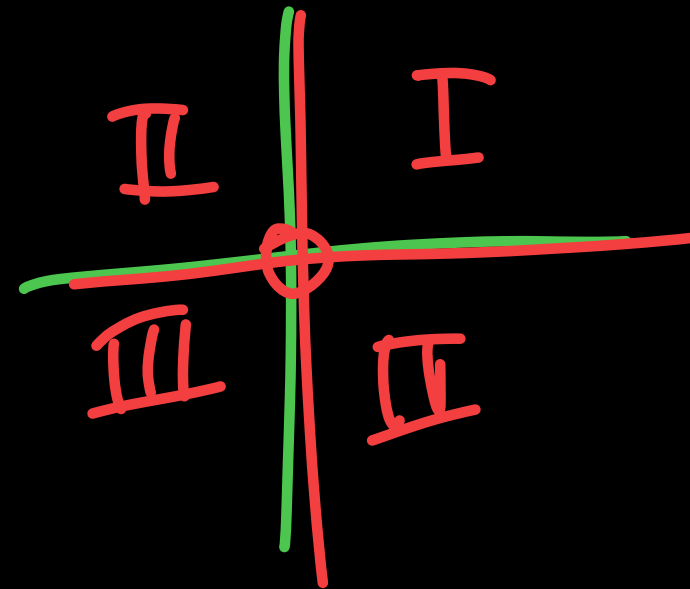
Karthkey if number is divisible by 5 only

Sakshi if number is divisible by 3 only

Anuj if number is divisible by 5 & 3 both

Hitesh if number is not divisible by 5 or 3

★ **HW:** Given a point (x, y) , write a program to find out if it lies in the 1st Quadrant, 2nd Quadrant, 3rd Quadrant, 4th Quadrant, on the x-axis, y-axis or at the origin.



Short Circuiting

false
↑
`if (c1 && c2)` → *ye check nahi hogi*

`if (c1 or c2)`
↓ ↓
true *ye check nahi hogi*

Nested If-Else

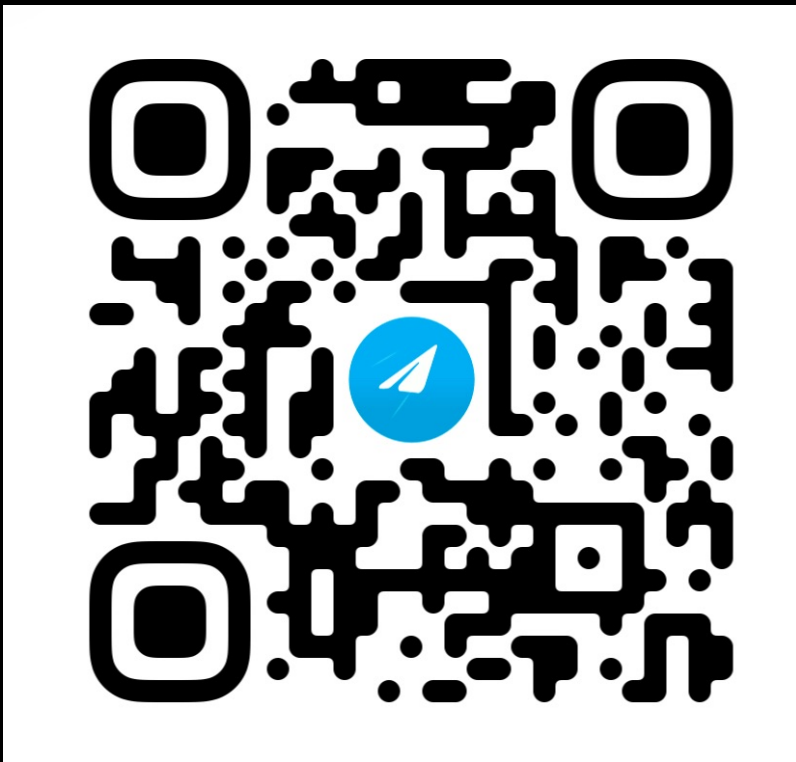
```
if ( ) {  
    if ( ) —  
    else —  
  
}  
else {  
    if ( ) —  
    else —  
  
}
```

Ques: Take 3 positive integers input and print the greatest of them. *a, b, c*

HW

~~Ques:~~ Take 3 positive integers input and
print the ~~greatest~~ of them. (nested)
least

HW: Take length and breadth of rectangle as input and write a program to find whether the area of rectangle is greater than its perimeter.



https://t.me/+YbgxkQd4_HcyNmM1